Janavahini: what it costs and how we get it to the ghats by June 2027

A pilot at one model compartment costs about **₹3.7–5.5 lakh** all-in. The full site costs **₹30–50 lakh** all-in if it reuses the CCTV the state is already installing, or **₹46–72 lakh** with our own cameras. That is under 0.02% of the ₹4,026 crore of Pushkaralu works. Every figure below comes from a public Indian price listing (November 2026) or a stated assumption.

Team IIIT Hyderabad: Divya Teja Yekkaluru, Pallamreddy Viswas Reddy

Challenge RTIH × RMC Innovation-Led Solutions for Godavari Pushkaralu 2027 **Date** November 2026

₹8.8–13.5k
one gate node: counters on 4 barricade lanes and a signal light

₹3.7–5.5 L
pilot all-in: 11 nodes, a gateway, operations and validation

₹30–50 L
full site all-in: 244 nodes, 10 gateways, reusing state CCTV

3.0–6.3 p
paise per expected pilgrim, Tier A (8–10 crore pilgrims) [11]

1 · WHAT WE ARE COSTING: THE SITE MODEL, NODE BY NODE

We cost exactly the site we simulated: 10 Nagaras feeding 10 approach roads, 20 junctions and 68 ghat compartments, each with its own gate and exit. A boundary narrow enough for barricade lanes (6 m or less) gets line counters; a wide area gets a density reading. This is the layout that kept 100 of 100 simulated surge draws safe.

Place	Width	Count	Sensing	Node type
Compartment gate	5 m	68	Line counters on 4 barricade lanes (1.2 m each)	Counter node · **136**
Compartment exit	3 m	68	Line counters on 2–3 lanes	
Ghat compartment	20 m	68	Density reading (camera)	Density node · **108**
Junction	25 m	20	Density reading	
Approach road	30 m	10	Density reading	
Nagara (holding area)	40 m	10	Density reading	
Zone gateway (one per Nagara zone)	–	10	Bridges the mesh to the ICCC over 4G	Gateway · **10**
Total		**254**	**244 nodes + 10 gateways**	

2 · BILL OF MATERIALS PER UNIT

Component	Gate node ₹	Exit node ₹	Density node ₹	Price basis
ESP32 board with LoRa radio (SX1276); runs the alert, the Kalman filter and the mesh	1,650	1,650	1,650	Hubtronics listing [1]

Component	Gate node ₹	Exit node ₹	Density node ₹	Price basis
VL53L1X time-of-flight sensors, a direction-aware pair per 1.2 m lane, shaded: 8 at a gate (4 lanes), 4–6 at an exit	4,530 – 5,320	2,260 – 3,990	–	₹566–665 each, FlyRobo quantity pricing [2]
I2C multiplexer (the sensors share one address) and lane wiring	300 – 600	300 – 600	–	Our estimate
Three-colour signal tower with buzzer, 24 V, IP65	630 – 3,135	630 – 3,135	630 – 3,135	Generic IndiaMART ₹630; Schneider XVGB3SH ₹3,135 [3]
IP65 enclosure, pole clamp, sun hood for the sensors	410 – 800	410 – 800	410 – 800	Enclosure ₹110–300 [4]; clamp and hood our estimate
Power: 12 V 7 Ah sealed battery (about 15 h backup), adapter, wiring	1,300 – 2,000	1,300 – 2,000	1,300 – 2,000	Exide 12 V 7 Ah ₹900–1,300 [5]; rest our estimate
Per node	**≈ 8,800 – 13,500**	**≈ 6,600 – 12,200**	**≈ 4,000 – 7,600**	
Own edge camera (density nodes, Tier B only): Raspberry Pi 5 (2 GB) + Camera Module 3 Wide + outdoor housing, power supply, SD card. A reading every 60–120 s meets our requirement, so the Pi's CPU is enough and no AI accelerator is needed.	–	–	+ 10,400 – 14,100	Pi 5 2 GB ₹5,145–7,287 [6]; camera ₹3,210 + GST [7]; housing and power our estimate
Zone gateway: ESP32 LoRa board (our gateway firmware) + 4G router + enclosure and power			5,800 – 7,800 each	TP-Link MR100 ₹2,690–3,699 [8]; 4G data about ₹400/month (our estimate)

The signal tower sets most of the range: a generic road-side unit at the low end, a branded industrial unit at the high end. The pilot will tell us which survives sun and rain. For comparison, one commercial dual-lens people-counting camera costs about US$764 [9]; our gate node, covering four lanes, costs ₹8,800–13,500.

3 · THE PILOT AND TWO FULL-SITE OPTIONS

Pilot · Kotilingala model compartment

₹3.7 – 5.5 lakh

6 counter nodes (3 gates, 3 exits), 5 density nodes (road, junction, 3 compartments) with **our own cameras**, 1 gateway. Own cameras let us measure camera accuracy against manual counts and the state's CCTV.

Tier A · reuse state CCTV

RECOMMENDED

₹30 – 50 lakh

The state is already installing CCTV at entry and exit points [11]. Density is computed from that CCTV and sent to the nodes. If the network fails, a wide-area node reports at least WATCH, which was safe in all 40 simulated camera-failure runs but means more holds.

Tier B · our own camera on every wide area

₹46 – 72 lakh

A camera is wired to each of the 108 density nodes, so wide-area density survives a network failure. Tier A plus own cameras only where CCTV does not reach falls between the two; the pilot data decides where.

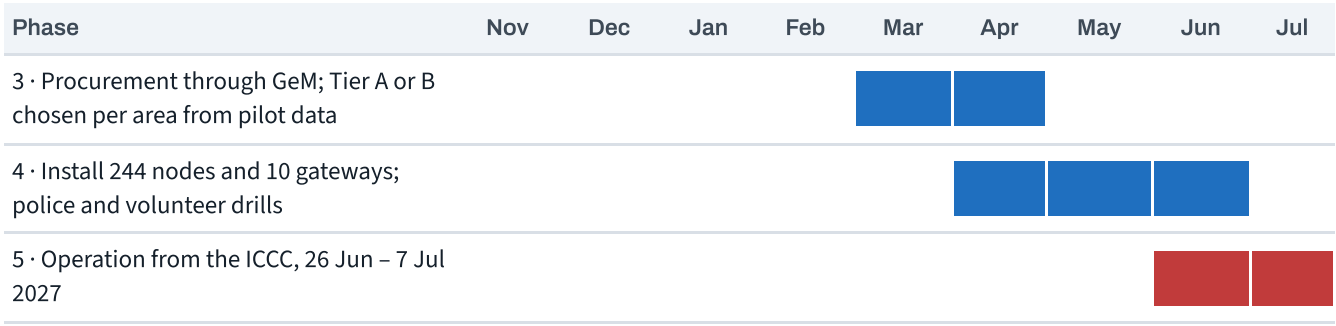
4 · INDICATIVE BUDGET (₹ LAKH, RANGES LOW – HIGH)

Line item	Pilot	Full site Tier A	Full site Tier B	Basis
Node and gateway hardware	1.24 – 1.93	15.4 – 26.4	26.5 – 41.6	Counts (§1) × unit costs (§2)
Spares	0.12 – 0.19	1.5 – 2.6	2.7 – 4.2	10% of hardware
GST	0.24 – 0.38	3.0 – 5.2	5.3 – 8.2	18% on hardware and spares; an upper bound, since some listed prices already include it
Installation	0.07 – 0.12	1.4 – 2.5	1.4 – 2.5	Two-person crew, 8 units a day, ₹1,000–1,500 per person-day (statutory floor for skilled work ≈ ₹520/day [10]); ₹300–600 of wiring per unit
4G data for gateways	0.04	0.36	0.36	₹400 a month × 9 months
Field operations	1.12 – 1.58	5.2 – 7.6	5.2 – 7.6	Pilot: 4 technicians × 3 days; our team of 4 × 10 days. Event: 20 technician shifts a day × 16 days (setup and the 12 days); our team of 4 on call at the ICCC for 20 days at ₹2,500–3,500 per person-day for travel and stay
Validation, drills and training	0.50 – 0.80	0.5 – 1.0	0.5 – 1.0	Pilot: manual-count validation and a bench test kit. Event: Telugu/English protocol cards and two drills with police and volunteers
Contingency	0.33 – 0.50	2.7 – 4.6	4.2 – 6.6	10%
Total	**≈ 3.7 – 5.5**	**≈ 30 – 50**	**≈ 46 – 72**	
Software licences	0	0	0	Open-source firmware, backend and dashboard
Server		ICCC-provided		One server or VM in the RMC or state data centre
Density from CCTV (Tier A)		ICCC analytics, or ≈ 1.5 – 2.5 extra		If the ICCC's video analytics cannot give a density per area, one GPU workstation runs an open crowd-counting model on the CCTV feeds (our estimate; not in the totals)

Indicative only. Prices are public retail listings from November 2026, before bulk discounts; for the full site they will be replaced by quotes or Government e-Marketplace (GeM) prices. Our team's development time is contributed in kind. Labour and travel rates are our estimates, shown so they can be checked. Memory shortages have raised single-board-computer prices in 2026, which affects Tier B most.

5 · IMPLEMENTATION TIMELINE (ALIGNED WITH CHALLENGE §11)

Phase	Nov	Dec	Jan	Feb	Mar	Apr	May	Jun	Jul
0 · Bench-tested boards (firmware builds and matches the simulator); demo on 24 Nov	■								
1 · Route map and pilot install at the Kotilingala model compartment; connect CCTV and ICCC	■	■	■						
2 · Live pilot at Maha Shivaratri; go / no-go review (§6)				■	■				



Nov 2026 – Jul 2027. Blue: build and scale · amber: pilot · red: Pushkaralu operation.

6 · GO / NO-GO BEFORE FULL-SITE SPENDING

Most of the money is spent only after the pilot passes these six checks. Each threshold comes from what the simulations needed.

1. **Counters:** within ±5% of manual counts at the pilot gates and exits. The Kalman filter was tested with 2–5% counter bias.
2. **Camera density:** a reading at least every 2 minutes with 20% error or less, measured against manual counts. This is the requirement from our sensitivity runs.
3. **Radio:** at least 70% of mesh messages delivered. The simulations assumed 30% loss.
4. **Uptime:** every silent node reported to staff; node uptime at least 99% over the pilot days.
5. **Response time:** in drills, the police at a road head act within 60 s of their light changing. In our field simulation (E14), with 70% of the crowd complying, up to 60 s of delay stayed safe in 20 of 20 surge draws; 2–3 minute delays kept only 14 and 9 of 20 safe.
6. **Police review:** warning lead time and the number of false alerts per day accepted by the police in charge of the pilot.

If a check fails, we fix the cause and re-test before buying for the full site; the pilot hardware is not wasted, because it becomes part of the full site.

7 · PHASES, DELIVERABLES AND WHAT WE MEASURE

When	What we deliver	Measured
Nov 2026	Firmware built for all three board types and bench-tested; live demo of network-down and dead-node behaviour	Mesh delivery rate, decision match with simulator
Nov 2026 – Jan 2027	Route map for the Kotilingala model compartment; 11 nodes and a gateway installed; CCTV and ICCC connected	Counting accuracy vs manual counts
Feb – Mar 2027	**Live pilot at Maha Shivaratri**; go / no-go report	The six checks in §6
Mar – Jun 2027	GeM procurement; 244 nodes and 10 gateways installed; Telugu/English staff messages; drills on the three-colour protocol	Drill response times, install acceptance tests
26 Jun – 7 Jul 2027	12-day operation from the ICCC, with our team on call	Live KPIs, incident log
After the event	Handover to RMC; nodes reused for Maha Shivaratri, Dasara, temple queues, bus stands and railway stations	Post-event report

Running cost after the event is small: 4G data for the gateways and a battery replacement every 2–3 years (₹900–1,300 per node).

8 · SUPPORT REQUIRED FROM THE ORGANISERS AND GOVERNMENT STAKEHOLDERS

1. Kotilingala model-compartment layout, planned pedestrian routes and ghat capacities, to build the route map.
2. Access to the CCTV feeds at entry and exit points and to the ICCC/RTGS integration interface. Tier A depends on this.
3. Permission and police coordination for a live pilot at Maha Shivaratri (Feb–Mar 2027), including mounting points and power at gates and junctions.
4. Power points at node locations, laid with the event's temporary lighting.
5. Police and volunteer time for drills on the three-colour protocol.
6. A server or VM in the RMC or state data centre.
7. Procurement of 244 nodes and 10 gateways through GeM or the ICCC system integrator.

9 · RISKS AND MITIGATIONS

Risk	Mitigation
Results so far are from simulation	The go / no-go checks (§6) at the Maha Shivaratri pilot come before most of the money is spent
Two neighbouring boards failing together	The ICCC covers the gap while the network is up; staff are alerted to every silent node
Precautionary slowing in near-misses (about 1 in 4)	Thresholds tuned on pilot data
Power, weather and theft	IP65 enclosures, a battery giving about 15 h of backup, mounting on existing barricades and poles; 10% spares on site
Counting on wide roads; counter drift	Counters only where barricades form lanes; camera density on wide areas; a Kalman filter on each node removes counter drift (simulated: 100 of 100 surge draws safe)
CCTV density unavailable or too poor (Tier A)	Our own cameras at those areas (Tier B unit cost: ₹10,400–14,100 each); the node reports at least WATCH while blind
Police act late, or crowds ignore the lights	Simulated (E14): with full compliance safe even at 3 minutes' delay; with 70% compliance safe in 20 of 20 surge draws up to 60 s, 9 of 20 at 3 minutes; drills train acting within a minute, and the light stands at the road head where the police are
Price changes before procurement	10% contingency; GeM rates; the signal tower and camera have alternative suppliers

SOURCES (ACCESSED NOVEMBER 2026)

1. Hubtronics, LILYGO LoRa32 ESP32 SX1276. hubtronics.in/ttgo
2. FlyRobo, VL53L1X module (₹663; ₹566 at 15+). flyrobo.in/vl53l1x-laser-time-flight-ranging-sensor-module
3. IndiaMART tower-light listings (3-tier, 24 V, IP65, buzzer, ₹630). m.indiamart.com/impcat/tower-light.html; Schneider Electric India e-shop, XVGB3SH (₹3,134.72). eshop.se.com/in
4. IndiaMART ABS enclosure listings (IP65). m.indiamart.com/impcat/abs-enclosure.html
5. Exide 12 V 7 Ah SMF (₹900; MRP ₹1,280). exidebatterywala.com; ₹1,300 at myorderstore.com
6. Raspberry Pi 5 2 GB: ₹5,145 at atomlabindia.com; ₹7,287 at robocraze.com
7. Raspberry Pi Camera Module 3 Wide (₹3,210 + GST). silverlineelectronics.in
8. TP-Link TL-MR100 4G router: ₹2,690 at golchhacomputer.com; ₹3,599 at pricebefore.com
9. Hikvision DS-2CD6825G0/C-IVS dual-lens people-counting camera (US$763.80). getic.com
10. Andhra Pradesh minimum wages, Shops and Establishments, from 1 April 2026: skilled ₹13,467/month (Zone I), Notification G/3186486/2026. dcpsol.com
11. The Hindu, "AI to help manage crowd during 2027 Godavari Pushkaralu in Andhra Pradesh", 25 Aug 2026: 8–10 crore devotees expected; 4,932 works costing ₹4,026 crore; CCTV at entry and exit points. thehindu.com